

MMR modification

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Equation

Original MMR

$$\text{mmr}(s_i) = (1 - \lambda) \delta_{\text{sim}}(s_i, q) + \frac{\lambda}{|R|} \sum_{s_j \in R} \delta_{\text{div}}(s_i, s_j)$$

Our modification

$$\text{mmr}_{\text{seq}}(i) = (1 - \lambda) \text{Rel}_G(i) + \lambda \left[(1 - \alpha) \text{Div}_{\text{curr}}(i) + \alpha \text{Div}_{\text{hist}}(i) \right]$$

Normalized group scores
from SIAA

$$\text{Div}_{\text{curr}}(i) = \frac{1}{|R_t|} \sum_{j \in R_t} \delta_{\text{div}}(i, j) \quad \text{Div}_{\text{hist}}(i) = \frac{1}{|H_t|} \sum_{h \in H_t} \delta_{\text{div}}(i, h)$$

In code we use (1-lambda) instead because
we implemented diversification as
negative max similarity

Parameters

- Alpha – current diversity vs history diversity
- Lambda – relevance vs diversification ratio
- For $\alpha = 0$ we get normal MMR – diversify only current round
- With $\alpha > 0$ we penalize based on history recommendations

Embeddings from genres

- We create a set of all genres $S = [\text{scifi}, \text{western}, \text{comedy}, \dots]$
- Then we one hot encode them so:
 - Sci-fi = $[1, 0, 0, \dots]$
 - Western comedy = $[0, 1, 1, \dots]$
- And use L2 normalization:
 - $V = V / ||V||$
- We then use cosine similarity for comparisons

Why it works

- We use relevance from SIAA which takes amount of satisfaction and dissatisfaction of others – every recommendation is meant for group, not for a single user
- Sequential diversity – with current diversity we make sure that in one round there are no similar recommendations, with history diversity we don't cycle one type of genres over and over again
- Easy to plug into sequential recommendations